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Title: API Integration and JSON Handling

Aim

To fetch data from a web API, process JSON data, and display the information using Python.

Software Requirements

- Python 3.12.5
- Windows OS
- VS Code / Notepad
- Requests Library

Program

```
import requests # Import requests module for API calls
try:
    # API URL
    url = "https://jsonplaceholder.typicode.com/users"

    # Send GET request to API
    response = requests.get(url)

    # Convert JSON response to Python data
    data = response.json()

    # Display user details
    for user in data:
        print("Name:", user["name"])
        print("Email:", user["email"])
        print("-----")

except requests.exceptions.RequestException as e:
    # Handle API-related errors
    print("API Error:", e)
```

Explanation

- Imports the requests module to communicate with APIs.
- Sends a GET request to retrieve user data.
- Converts JSON response into Python objects.
- Displays user names and email addresses.
- Uses exception handling to manage API errors.

Output

```
Microsoft Windows [Version 10.0.26200.8457]
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C:\Users\VICTUS\OneDrive\Desktop\Task2_API_Integration>python api_data.py
Name: Leanne Graham
Email: Sincere@april.biz
-----
Name: Ervin Howell
Email: Shanna@melissa.tv
-----
Name: Clementine Bauch
Email: Nathan@yesenia.net
-----
Name: Patricia Lebsack
Email: Julianne.OConner@kory.org
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Name: Chelsey Dietrich
Email: Lucio_Hettinger@annie.ca
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Name: Mrs. Dennis Schulist
Email: Karley_Dach@jasper.info
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Name: Kurtis Weissnat
Email: Telly.Hoeger@billy.biz
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Name: Nicholas Runolfsdottir V
Email: Sherwood@rosamond.me
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Name: Glenna Reichert
Email: Chaim_McDermott@dana.io
-----
Name: Clementina DuBuque
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```

Conclusion

The program successfully fetched data from an API, processed JSON data, and displayed the required information using Python.